CS4.406 Term Project — How Fast Does a News Recommender Go Stale?

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Summary

We froze five scorers (word matching BM25, frozen multilingual MiniLM meaning matching, co-click history, recency, popularity) plus a random shuffle on 1–14 days of past clicks, and scored real later clicks against the articles published in the 48 hours before each click. The gap to a model rebuilt the day before is the *decay*. All intervals are 95% paired bootstrap intervals over clicks; n is given with every table. Main results, EB-NeRD (Danish, 14 fit days, 105 cells):

- **Click history and popularity fail by coverage.** One hour after a fit the click-history model can score 79% of clicks; after 6 h 19%, after 24 h 0.9%; after 3 days none. New articles have no clicks yet, so there is nothing to compute.
- **Word and meaning matching fail through the user profile, not the articles.** With a profile frozen at the fit date, word matching loses 0.0030 $[0.0024, 0.0036]$ MRR by day 7 (10% of 0.0299) and 0.0059 by day 13. With a live profile (last 50 clicks before the click) the loss is about 0.0001.
- **Recency is far ahead** (top-10 hit rate 0.515 $[0.513, 0.517]$) and never goes stale. Content signals are only 2–3 times the shuffle.
- **Rebuild frequency.** Click history must be rebuilt at least daily, and hourly on a site like EB-NeRD. Word matching with a frozen profile loses 12% of its MRR if rebuilt once per 14 days instead of daily; with a live profile a rebuild buys nothing. Meaning matching never needs a rebuild for decay.
- The **turnover** differs sharply by site: the share of new clicks that go to articles unseen by a frozen model at 6 h is 90.8% on EB-NeRD, 48.5% on Adressa and 15.5% on NPR.

1 Question and setup

Click-forward evaluation. For a click (u, a, t) the candidate pool is every article published at or before t and within the previous 48 h. The scorer ranks the clicked article a in that pool. A model *frozen* at fit date r is fitted on the 7 days of clicks ending on r and scores clicks on later days d; *lag* is $d - r$. The *control* for day d is the model fitted on $d - 1$ (lag 1). **Decay = control − frozen on the same clicks.** The 14 fit dates are 10–23 May 2023 and the evaluation days 11–24 May; the sample is 20,000 clicks per day chosen by keyed hash (seed 20261029).

Two profile regimes. In regime A the user profile is frozen with the model (clicks up to the fit). In regime B it is live (the last 50 clicks strictly before t). Click history and popularity use no profile of their own, so A and B differ only for the content signals and click history's profile.

Already-read rule. A click whose target is in the user's own earlier clicks (cap 300) is a re-read and is left out; the user's history is removed from every pool. The rule is identical for every cell, scorer and regime, so all comparisons stay paired.

Metrics. MRR and recall@10 (the top-10 hit rate) are primary; AUC and nDCG@5 are in the result files. Candidates a scorer cannot score tie at the bottom with a keyed-hash tie-break. *Coverage* is the share of target articles the frozen model can score. Because the negatives are the other pool articles (weak negatives, not impression slates), absolute levels are low; the decay gap is the object of study.

Fresh versus decayed. Targets are split into S0 (published before the fit end), S1 (after the fit, before day d) and S2 (published on day d). The stratum terms sum exactly to the gap (asserted in the analysis).

Bootstrap. One multinomial resample-count matrix per evaluation day ($B = 1000$, seed from `SeedSequence([20261029, day])`) is shared by every scorer and fit date, so a decay is a paired difference. Days enter a lag's average with equal weight; the days themselves are taken as given.

2 Data

	EB-NeRD (main)	**NPR** (second)	**Adressa** (not run)
Publisher, language	Ekstra Bladet, Danish	Globo, Portuguese	Norwegian news site
Days of clicks	22 (May 2023)	118 (Jan–Apr 2023)	90 (Jan–Mar 2017)
Clicks, users	113.4 M, 791,582	17.6 M, 1,162,802	35.2 M, 3,614,911
Typical age of a clicked article	1.4 h	13.8 h	5.8 h
Evaluation here	14 days, 20,000 clicks a day	14 days (11–24 Mar), all impression clicks	no (turnover only)

For EB-NeRD we use the click history of the validation users, because its volume and type of click are constant across dates. 27.2% of the 4,468 EB-NeRD articles published 4–25 May were never clicked in the corpus (see the sensitivity run, Section 4.6).

3 Pipeline and checks

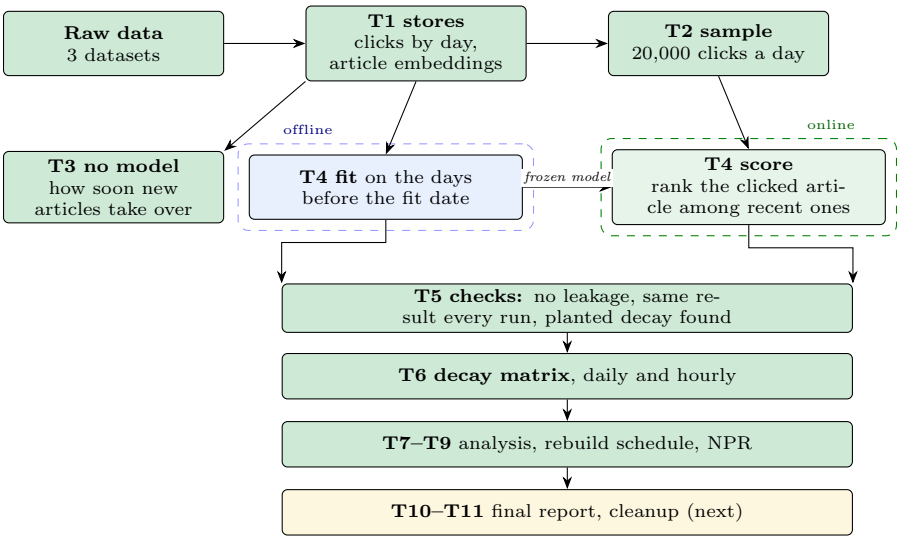


Figure 1: The pipeline. Green is done, yellow is next. A monitor wraps every unit and watches time, memory and disk.

The determinism contract: keyed randomness, one unit writes its own file, one numeric thread per worker, sorted outputs, write to a temporary file then rename, a manifest line per unit with a sha256 of every file. The T5 gate (`results/metrics/t5_gate.txt`) checks that 1 worker, 2 workers, reversed order and a resume produce identical hashes; that recency has a gap of exactly 0; that the stratum terms sum to the gap; that a pool allowing post-click articles fails (mutation test); that day edges are right for each dataset; and two planted worlds: with article turnover the click-history gap is recovered (0.0348 ± 0.0049 at $n = 5000$) and with zero turnover every gap lies within 3.5 standard errors (largest 1.87). 33 unit tests and 8 gate tests pass.

4 Results

4.1 Turnover without any model

Share of clicks at h hours after a fit that go to articles published after the fit (13 fit days for EB-NeRD, 80 for Adressa, 85 for NPR; bootstrap over fit days). n clicks at 6 h: 3.46 M, 1.26 M, 72,136.

	1 h	3 h	6 h	12 h	24 h
EB-NeRD	19.5 [14.6, 23.9]	77.2 [72.7, 81.1]	90.8 [88.5, 92.5]	95.4 [94.8, 95.8]	96.2 [95.8, 96.5]
Adressa	6.4 [4.8, 8.3]	24.8 [21.6, 28.1]	48.5 [45.4, 51.5]	73.2 [71.3, 75.1]	87.8 [86.8, 88.7]
NPR	1.0 [0.7, 1.5]	3.2 [2.4, 4.2]	15.5 [11.3, 20.1]	49.8 [41.0, 58.2]	85.6 [80.6, 89.7]

This depends on the hour of the fit: an evening or midnight fit stays stale longer (EB-NeRD hours to 50%: 6 for a midnight fit, 2 at 06:00). EB-NeRD times are read as UTC; this is not confirmed.

4.2 Daily decay, EB-NeRD

MRR (levels and decay with 95% CIs), regime A (frozen profile) and B (live profile). Clicks scored: 236,387 at lag 1, 202,331 at lag 3, 135,583 at lag 7, 34,479 at lag 13 (fewer fit-to-day pairs exist at long lags; lag 13 is a single cell).

		lag 1	lag 3	lag 7	lag 13
Word, A	level	0.0310 [0.0306, 0.0314]	0.0309 [0.0305, 0.0313]	0.0299 [0.0295, 0.0305]	0.0248 [0.0239, 0.0258]
	decay	0	0.0008 [0.0004, 0.0012]	0.0030 [0.0024, 0.0036]	0.0059 [0.0047, 0.0071]
Word, B	level	0.0325 [0.0321, 0.0329]	0.0331	0.0341 [0.0336, 0.0347]	0.0313
	decay	0	–	0.00013 [0.00007, 0.00019]	–
Meaning, A	level	0.0250 [0.0247, 0.0253]	0.0247	0.0240 [0.0236, 0.0244]	0.0212 [0.0204, 0.0221]
	decay	0	-0.0002 [-0.0004, 0.0001]	0.0007 [0.0003, 0.0011]	0.0013 [0.0005, 0.0022]
Meaning, B	level, decay	0.0259, 0	0.0253, 0	0.0253, 0	0.0233, 0
Shuffle	level	0.0172	0.0177	0.0176	0.0150
Recency	level	0.190	0.195	0.196	0.179

Meaning B is exactly 0 because the article embeddings do not depend on the fit and the profile is rebuilt from clicks, so the frozen and control models are the same function. The meaning A loss at lag 3 is within noise.

Where the loss sits. The stratum split for word, regime B, shows nearly all of the (small) decay comes from articles published on the evaluation day: at lag 7 the S2 term is 0.00012 of the total 0.00013, and S1 contributes 0.000006. The content signals lose accuracy on the newest articles, not on old ones.

Click history and popularity. Overall MRR is penalised by cold start: an unscorable target ranks below scorable older articles, so these two score *below* the shuffle (0.0079 [0.0077, 0.0080] for click history at lag 1, against shuffle 0.0172), and their "decay" is negative at lag ≥ 3 because the frozen model falls back to shuffle level. Coverage is

therefore the honest measure: 12.0% at lag 1, 0.4% at lag 2, 0 from lag 3. On the clicks it *can* score, accuracy is fine and the comparison with the control is not significant (lag 2 click history: 0.139 against 0.141, decay 0.002 [-0.015, 0.020], 12 cells, but only 0.4% of clicks).

4.3 Hourly fast axis, EB-NeRD

Fits every 3 h on 17–21 May, each scoring the next 24 hourly buckets of 5,000 clicks (17–22 May; 40 cells per offset, 144 units), regime A. Each hour is also the control of its own bucket.

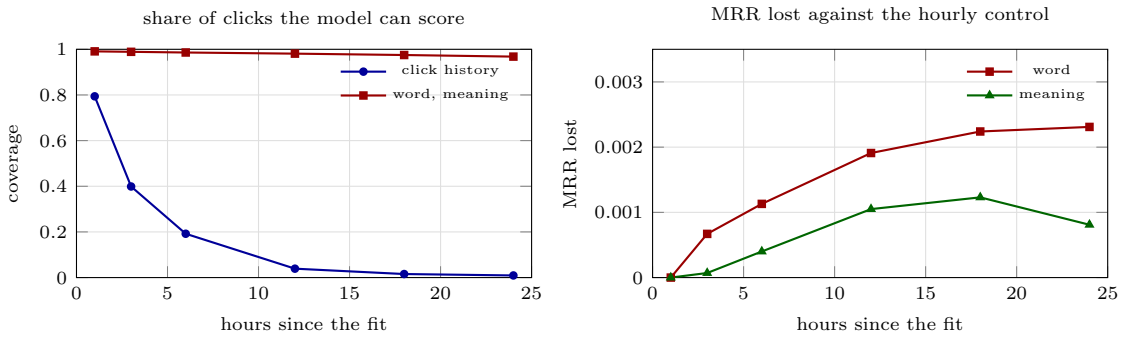


Figure 2: Hourly axis, EB-NeRD, regime A, 40 cells per point. Intervals are in the table below and `results/rq_hourly/decay_by_hour.csv` (all are narrower than the markers except at 3 h for meaning).

	3 h	6 h	12 h	24 h
Word, MRR lost	0.00067 [0.00052, 0.00082]	0.00113 [0.00093, 0.00133]	0.00191 [0.00163, 0.00219]	0.00231 [0.00196, 0.00264]
Meaning, MRR lost	0.00007 [-0.00001, 0.00015]	0.00040 [0.00028, 0.00052]	0.00105 [0.00088, 0.00122]	0.00081 [0.00056, 0.00105]
Click history coverage	0.399 [0.396, 0.401]	0.192 [0.191, 0.194]	0.039 [0.038, 0.040]	0.0093 [0.0088, 0.0097]
Click history, recall@10 lost on clicks it can score	–	0.059 [0.055, 0.062]	–	0.009 [-0.009, 0.028]

Word matching at 24 h is 0.0023/0.0322 = 7% of its level. The click-history loss on the clicks it can still score is large at 6 h (recall@10 0.045 against 0.104 for the control) but undetermined at 24 h, where only 0.9% of clicks are scorable.

4.4 Rebuild schedule (RQ3)

The replay uses the same cells: with rebuild interval k days, day d uses the latest rebuild at or before $d-1$. MRR over the 14 days, with the total fit seconds (mean of the measured fit time per scorer).

Interval (days)	1 (14 rebuilds)	7 (2 rebuilds)	14 (1 rebuild)
Word, A (frozen profile)	0.0310 [0.0306, 0.0314]	0.0298 [0.0295, 0.0302]	0.0274 [0.0270, 0.0277]
Word, B (live profile)	0.0325 [0.0321, 0.0329]	0.0324 [0.0320, 0.0328]	0.0324 [0.0320, 0.0327]
Meaning, A and B	identical at every interval		
Click history, B	0.0079 [0.0077, 0.0080]	0.0148 [0.0146, 0.0151]	0.0161 [0.0158, 0.0163]
Fit time, total	306 s (click history), 29 s (word)	44 s, 4 s	22 s, 2 s

The rising click-history MRR with the interval is a coverage artefact: an older, wider window reaches more of the older articles in the pool, and the shuffle-level score of unscorable targets rises as fewer targets are cold. It is not evidence that a stale click-history model is better. The cost of one click-history fit is 22 s and a peak near 7 GB (two workers fit in the 8 GB cap we used). A recency-backed hybrid would change this table and was not built.

4.5 Second dataset: NPR

14 fit days (10–23 March 2023) and evaluation days (11–24 March); all impression clicks, 119,172 at lag 1. The pool is about 1,500 articles (a typical clicked article is 13.8 h old). Regime A, lag 1: MRR popularity 0.162 [0.161, 0.163], click history 0.105 [0.104, 0.107], word 0.0181 [0.0175, 0.0187], meaning 0.0179 [0.0174, 0.0185], recency 0.0066 [0.0065, 0.0068], shuffle 0.0058 [0.0056, 0.0060]; recall@10 popularity 0.550 [0.547, 0.553], click history 0.249 [0.246, 0.251].

- The ranking of the signals differs from EB-NeRD: on NPR, popularity and click history are the best, and recency is no better than the shuffle. Coverage is high (97% at lag 1, 96% at lag 7 for click history; word and meaning 87% at lag 1 and 73% at lag 7, from users with no profile).
- Word matching loses 0.0012 [0.0010, 0.0015] MRR at lag 3 and 0.0024 [0.0020, 0.0029] at lag 7 (14% of its 0.0167 level), meaning 0.0022 [0.0017, 0.0027] at lag 7.
- **Unexplained:** popularity gets *better* with an older fit (MRR −0.112 [-0.113, -0.111] "decay" at lag 3; recall@10 −0.084). We have not found the cause; it is reported as a measurement only. NPR needs a position-based baseline we did not build.

4.6 Sensitivity: without the never-clicked articles

All 1,216 articles (27.2% of 4,468) with no click anywhere in the 22-day corpus were removed from every pool (oracle: it uses future clicks). The evaluated clicks are identical (for example 17,037 clicks in pool on the first fit in both runs). Regime B, lag 1, recall@10:

	Recency	Shuffle	Meaning / Word
All articles in the pool	0.515 [0.513, 0.517]	0.027 [0.026, 0.027]	0.047 / 0.061
Never-clicked removed	0.634 [0.632, 0.636]	0.040 [0.039, 0.041]	0.062 [0.061, 0.063] / 0.063 [0.062, 0.064]

Every level rises, but the decay does not change: word B MRR lost at lag 7 is 0.00013 in the main run and 0.00010 [0.00003, 0.00016] here; meaning and recency stay at 0. Click history and popularity keep the same coverage (12% at lag 1, 0 at lag 7); their negative lag-7 "decay" grows only because the shuffle level they fall to rose. This is a ceiling on the levels, not a realistic setting and not a bound on decay.

4.7 Does decay move with daily volume? (exploratory)

Per evaluation day at lag 3, regime B, Spearman ρ against the articles published that day (152–266) and the clicks that day (4.2–6.8 M), bootstrap interval over the 12 days: word MRR lost against articles published -0.42 [-0.87, 0.24], against clicks -0.52 [-0.98, 0.20]; click-history MRR lost against articles $+0.43$ [-0.14, 0.81], against clicks -0.20 [-0.77, 0.55]. Meaning (always 0) and click-history coverage (always 0 at lag 3) are constant. **No relation is detectable** with 12 days; this needs a longer span or a second site and is not a finding.

5 What went wrong, and what we changed

- **Re-reads (the main error).** The first full run scored the user's already-read articles. A check of our own showed 15% of targets were already in the user's history and that already-read articles outranked targets, which also explained a "fresher profile scores lower" oddity. Word B MRR was 0.0573 with them and 0.0143 on the non-re-reads. We deleted every v1 result and reran under the already-read rule (decision D28).
- **Two hourly units marked complete with missing cell files** (19 files, fits 18 May 18:00 and 21:00). Found when the analysis failed; recomputed. The cause was not found. A manifest-against-files check belongs in the cleanup.
- **Memory.** Click history needs about 7 GB per fit, not the 1.5 GB planned; only two workers run at once.
- **Corpus.** The first plan assumed a uniform click corpus; EB-NeRD is not uniform, so we chose click-forward evaluation on one cohort.
- **Duplicates.** The first stores held 107,633 (NPR) and 7,693 (Adressa) duplicate rows, found by counting distinct keys and rebuilt.
- **Our own earlier summary was wrong** to say word matching "barely goes stale"; that held for the live profile only (corrected in Section 4.2).

The full record is in `log.md`: decisions D1–D28, findings P1–P47, an errata table with dates, and a session log.

6 Limitations, stated plainly

- One site (EB-NeRD) for the main result, one cohort (the validation users), one month, 14 fit days. The decay-by-volume question cannot be answered with this span. NPR is a second site; Adressa was not run beyond the turnover measure.
- Negatives are the other pool articles, not impression slates, so levels are low and overall MRR is penalised by cold start. We report coverage and accuracy on the scorable clicks beside it, but the team still has to choose the headline metric.
- The pool window is 48 h; sensitivity to it, to the sample size and to the seed was planned and not run. The 20,000 clicks per day are a sample of 4–7 M.
- EB-NeRD times are assumed UTC; the hour labels may be shifted by 1–2 h.
- The decay is measured only against a model rebuilt the day before (or the hour before); a rebuild every few minutes is not tested. Days are taken as given in the bootstrap, so the intervals do not cover day-to-day variation.
- No hybrid model (recency as the fallback for unscorable candidates) and no position-based baseline were built, so the rebuild-interval recommendation applies to each signal alone.
- Popularity improving with an older fit on NPR is unexplained.

7 Reproducing

From `code/` with the project's Python environment: `decay.py <root> --fits 2023-05-10 2023-05-23 --last 2023-05-24` (daily matrix), `decayh.py <root> --first 2023-05-17 --last 2023-05-22 --fit-days 2023-05-17 2023-05-21 --step 3` (hourly), `analyze.py <root>` and `analyze_hourly.py <root>` (tables with CIs), `decay.py ... --name npr --out results/npr/decay_matrix` with `analyze.py --src results/npr/decay_matrix`, `decay.py ... --clicked-only --out results/sens_clicked_only/decay_matrix` (sensitivity), `volume_check.py <root>`, and `test_t5.py` for the gate. Data stays under `data/` (not in the repository). Run logs and the gate record are in `results/metrics/`. This repository is a temporary share for the team; the tarball of the Adressa data was never extracted.